Experiment No: 2

Course Outcome: CO2 **Blooms Level: L3**

Aim: To Collect data from Social Media Platform/s using tools

Abstract:

This experiment focuses on collecting business-oriented social media data from selected platforms such as YouTube, Google Reviews, Facebook. It involves connecting to these platforms and capturing relevant data using techniques like web scraping, crawling, and parsing. The collected data is structured and cleaned to make it suitable for further analysis and business intelligence tasks. This process helps organizations gain insights into customer opinions, market trends, and brand performance from social media content.

Sample Input and Output:

Case 1: Provide the YouTube video ID and API key to the YouTube Data API.
Output: Authentication is successful and the video resource is accessed.

Case 2: Execute the API request to fetch video comments.
Output: User comments along with author name and timestamp are retrieved.

Case 3: Store the extracted comments in structured format (CSV/JSON).
Output: YouTube comments are successfully saved for further analysis.

Theory:
Social media text analytics is a technique for extracting and analyzing business insights from the textual content of social media platforms. It can be used to understand emotions and sentiments expressed in user-generated content related to a brand or product. Text analytics has roots in various fields such as data mining and natural language processing. It is useful for quickly and accurately understanding the vast volume of social media content generated by users. Approximately 80% of data in organizations is textual, making text analytics an important tool for business intelligence.

Youtube Extension:
The YouTube Data API allows developers to programmatically access public YouTube data such as videos, channels, and comments. It requires an API key for authentication and provides structured data in JSON format. This enables efficient and scalable extraction of YouTube comments for analysis.

Chrome Extension:
A Chrome extension is a browser-based tool that adds extra functionality to the Google Chrome browser. It can extract visible web data such as reviews or comments directly from a webpage without complex programming. Chrome extensions are easy to use and suitable for quick data collection tasks.

Program:

Part 1- Scraping Youtube Video Comments

Step1: Get API Key –
Go to the *Google Cloud console • (Google Cloud Consolehttps://console.cloud.google.com) and sign in with your Google Account
Step2: Click the project drop-down menu in the top bar and select or create the project you want to use.

Steps3: Click the hamburger menu in the top left and select APIs & Services >Select Youtube Data V3 API> Credentials.
Steps4: Click the Create credentials button and select API key.
The API key will be displayed in a pop-up window.
Click the COPY button to copy the API key to your clipboard.
Steps5: You can use the API key in your application to access the Google Cloud APIs. Be sure to keep the API key confidential, as it can be used to access your Google Cloud resources.

Extracting Youtube comments from given video and extracting comments to perform sentiment analysis

In [ ]:
```
import requests
```

In [ ]:
```
video_id ="yTMYtcQLLaw"
```

In [ ]:
```
api_key = "AIzaSyB1LxwOqpakEJv_J3dWbt_PQHd5GK-5Gdc"
```

In [ ]:
```
video_info_url = f"https://www.googleapis.com/youtube/v3/videos?part=snippet&id={video_id}&key={api_key}"
video_info_response = requests.get(video_info_url)
video_info_data = video_info_response.json()
```

In [ ]:
```
print(video_info_data)
```

Retrieve Youtube Video Comments

In [ ]:
```
comments_url = f"https://www.googleapis.com/youtube/v3/commentThreads?part=snippet&videoId={video_id}&key={api_key}"
comments_response = requests.get(comments_url)
comments_data = comments_response.json()
```

In [ ]:
```
comments_data
```

In [ ]:
```
#extract the comment

comments = [item["snippet"]["topLevelComment"]["snippet"]["textOriginal"] for item in comments_data["items"]]
```

In [ ]:
```
comments
```

In [ ]:
```
print(comments)
```

In [ ]:
```
import pandas as pd
```

In [ ]:
```
df = pd.DataFrame(comments)
```

In [ ]:
```
df.to_csv("output1.csv", index=False)
```

Output:

Comments Extracted from Youtube Video:


```
Out[7]: {'kind': 'youtube#commentThreadListResponse',
 'etag': 'Kzg_QDc-wwvKqBlDgcWPsMjaSgk',
 'nextPageToken': 'Z2V0X25ld2VzdF9maXJzdC0tQ2dnSWdBUVZGN2ZST0JJRkNKMGdHQUVTQlFpSklCZ0FFZ1VJaUNBWUFCSUZDSWNnR0FBU0JRaW9JQm
dBSWc0S0RBaU1qOFBLQmhEQWpjNmFBdw==',
 'pageInfo': {'totalResults': 20, 'resultsPerPage': 20},
 'items': [{'kind': 'youtube#commentThread',
   'etag': 'Dm92aF2FZklmqaZxyllzmlG2y5Y',
   'id': 'UgyxI49iMHmyuvs75Ph4AaABAg',
   'snippet': {'channelId': 'UCnC8SAZzQiBGYVSKZ_S3y4Q',
    'videoId': 'yTMYtcQLLaw',
    'topLevelComment': {'kind': 'youtube#comment',
     'etag': 'f8gf1COsPbxkz2xirAModHA64rc',
     'id': 'UgyxI49iMHmyuvs75Ph4AaABAg',
     'snippet': {'channelId': 'UCnC8SAZzQiBGYVSKZ_S3y4Q',
      'videoId': 'yTMYtcQLLaw',
      'textDisplay': '<a href="https://www.youtube.com/watch?v=yTMYtcQLLaw&amp;t=4622">1:17:02</a>',
      'textOriginal': '1:17:02',
      'authorDisplayName': '@AbhishekTATIKONDA',
      'authorProfileImageUrl': 'https://yt3.ggpht.com/4Nb8sxJ2SG1SVW87vOVh47tytHqhpTrFWqinrE0XZ9ZvMle4jhXsHdflqGoIfmBNsVM
```

```
In [9]:  #extract the comment

         comments = [item["snippet"]["topLevelComment"]["snippet"]["textOriginal"] for item in comments_data["items"]]

In [10]: comments

Out[10]: ['1:17:02',
 'Why not SPEAK IN ENGLISH.\n\nModi's English is perfect.',
 'AAP aam kaise khate hai. Wahi Wala seen chahiye tha 😂😂😂😂😂😂.',
 'And now tell me his age',
 'please podcast with amit shah',
 'What is scripted interview....charan sparsh kr lete',
 'Feku Number 1 Of The World 🌍 \n\nZumlebaaz Of The World \n\n🤡🤡🤡🤡🤡🤡',
 'Manonio joshoshi Pradhan mantri shree Narendra Modi ji up ko chorono me Mera hriday se pranam or dhonyobad korti hu sir. U
p por hum 14o core desh bashio gorbit hu sir. Up sawal jobab bahuti amulyo or mohotyopurna hai sir. Nikhilesh kamath ji ko M
era hridai se pranam or new year ki avinondon korti hu sir. Up ko Mera new year ki hridai se koti koti pair, shubhokamona, d
honyobad, avinondon or nomon korti hu sir. Up ne khubi sadharon  balyo kal me bitaye tha lekin up ka koi ashor up ki jibon m
e nahi pore hai sir. Up ki vagyo up ko mukhya mantri cor tishri bar Pradhan mantri banaya hai. Up vogban ram ka rup hai sir.
Manusyo bankor aye hai. Isi liye desh ki sabi jonogon up ko itna sradha or pair korte hai sir. Shadhu ka rup up por birajman
hai isi liye itni shokti up ko milta hai sir. Up excellent ho, great ho, goribo ka vagban ho, narigon ke lie sanmanio ho, bi
r ho, or nomosyo ho sir. Up ka dil pe selut korti hu sir. Up ki ie nai soch or vogban ram jaisa bichardhara or rup bahuti op
urbo hai sir. Up ko mai abhar bakto korti hu sir. Joy shree ram. Joy shree radhe. ❤❤❤❤❤❤❤❤❤🙏🙏🙏🙏🙏🙏🙏🙏🙏🙏
🙏',
 '.\n\n\n
```

Conclusion:

In this experiment, data was successfully collected from social media platforms using both the YouTube Data API and Chrome extensions. The YouTube API enabled structured and reliable extraction of comments through authenticated access, while Chrome extensions provided a simple and user-friendly approach for collecting web-based reviews. Overall, these tools demonstrate effective methods for gathering social media data that can be further used for analysis and business insights.

Exercise 1:

To extract publicly available user reviews from Google using a Chrome extension, store the collected data in a structured format, for future analysis for useful business insights.

Abstract

This exercise focuses on extracting publicly available user reviews from Google using a Chrome extension. The collected data is stored in a structured format for ease of processing and analysis. The extracted reviews can be used to derive meaningful business insights and understand customer sentiment.

Exercise 2:

To extract publicly available posts, comments, and engagement data from social media platforms such as Facebook or Reddit using their respective official APIs, and store the collected data in a structured format for further analysis.

Abstract:

This exercise focuses on extracting publicly available data from social media platforms such as Facebook and Reddit using their official APIs. The collected posts and comments are stored in a structured format for efficient processing. This data can be analyzed to gain insights into user behavior and online discussions.